

PolyGlide 1226XF

Introduction

PolyGlide 1226XF is a micronized composite wax based on high density polyethylene (HDPE) that has been modified with two different types of high performance ceramic. The ceramic components significantly improve abrasion resistance and enhance the inherent lubricity of the HDPE without the use of PTFE.

PolyGlide 1226XF can be compared to MPP-611XF (same polyethylene wax without ceramic modification), and can also be evaluated as a PTFE-free alternative to Polyfluo 150:

Physical Property:	PolyGlide 1226XF	MPP-611XF	Polyfluo 150
Chemistry	HDPE/ceramic	HDPE	HDPE/PTFE
Mean particle size (µm)	3.5 – 5.5	4.0 – 6.0	3.5 – 5.5
Maximum particle size (µm)	15.56	22.0	15.56
Melting point (°C)	109 – 115	110 – 116	113 – 116
Density at 25 °C (g/cc)	0.99	0.96	1.04

Objective

Compare the performance of PolyGlide 1226XF vs. Polyfluo 150 for Taber abrasion and Coefficient of Friction (COF).

Test Methods

- ASTM D4060 Standard Test Method for Abrasion Resistance
- ASTM D1894 Standard Test Method for Coefficient of Friction

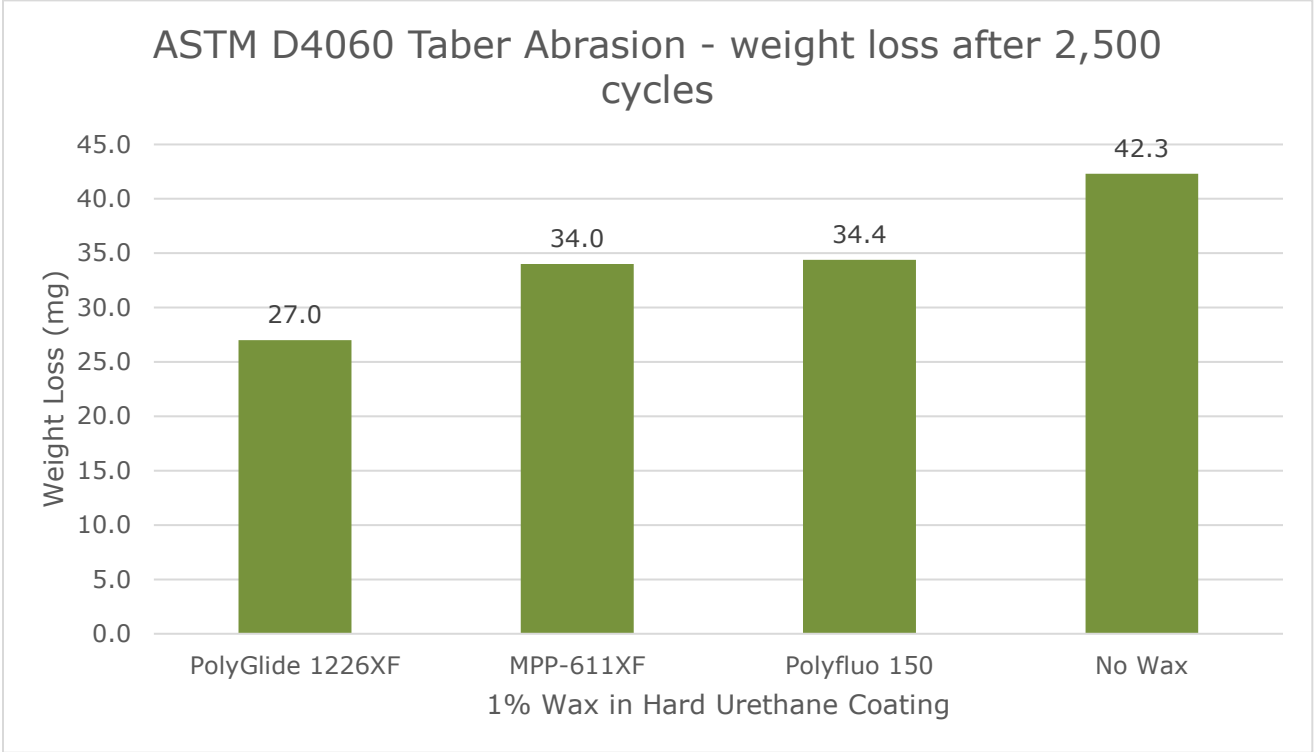
Formulations and Study Methodology

Hard Urethane Coating Formula:	Weight %:
Urotuf L63-MPW-38 (Reichhold)	97.5
Diethylene glycol monoethyl ether	2.0
BYK-345	0.2
TEGO Foamex 805 N	0.3
Total:	100.0
Nonvolatiles (Solids):	38.4

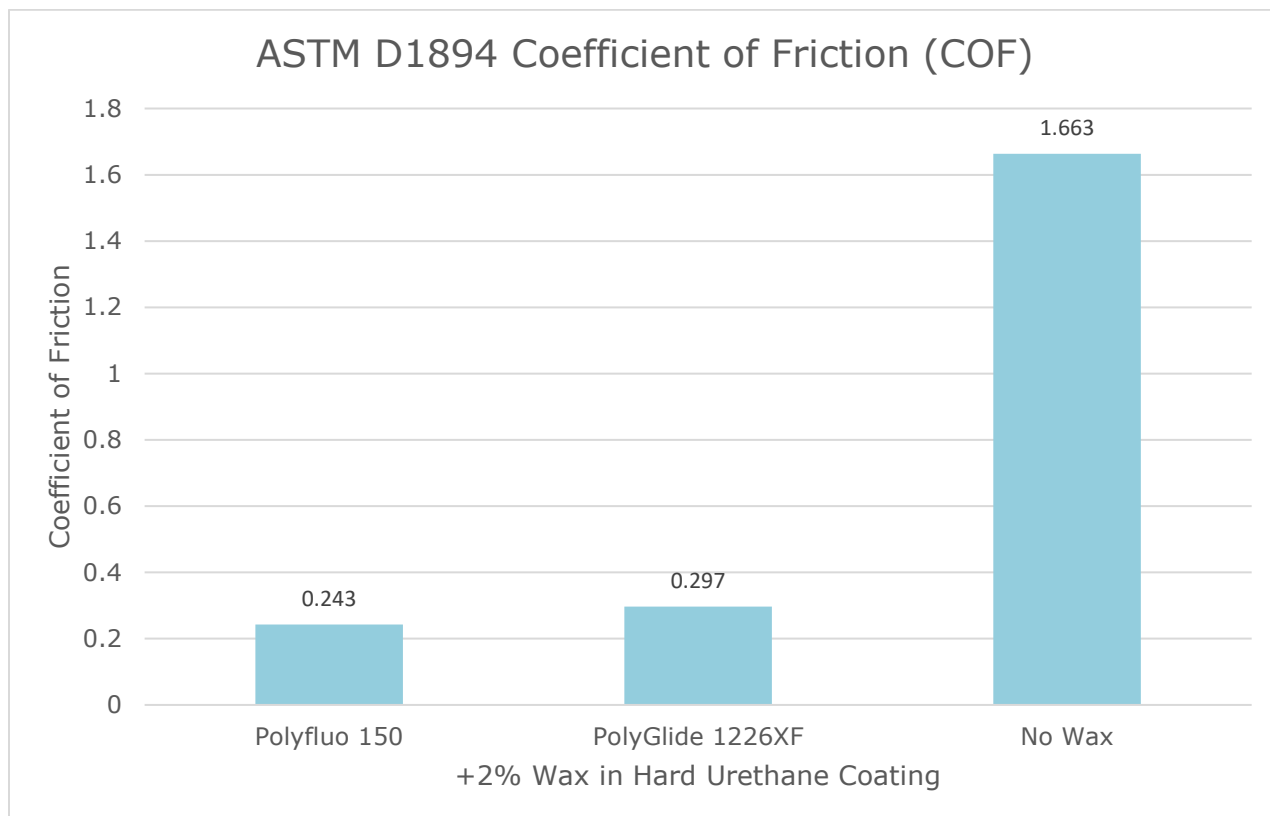
Sample Preparation

- Coatings for Taber Abrasion testing were drawn down with a #60 wire wound rod to produce approximately 2 mil (50 µm) DFT coatings.
- Coatings for COF testing were drawn down with a 3 mil (75 µm) Bird Film Applicator to produce approximately 1 mil (25 µm) DFT coatings.
- All coatings were allowed to cure for 7 days at room temperature at ~50% humidity.

Results



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Conclusions

PolyGlide 1226XF compares favorably to both MPP-611XF and Polyfluo 150:

- PolyGlide 1226XF has **21% higher abrasion resistance** vs. MPP-611XF
- PolyGlide 1226XF has **22% higher abrasion resistance** vs. Polyfluo 150
- PolyGlide 1226XF has comparable lubricity (coefficient of friction) vs. Polyfluo 150

PolyGlide 1226XF has an ultrafine maximum particle size (15.56 μm) and is therefore recommended as a replacement for PTFE based wax additives in can and coil coating applications.

For customers who are using Polyfluo 900, PolyGlide 1226XF can be considered as a PTFE-free alternative.